
PrimeCore Minerals

VLAN Design

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Introduction

The introduction of VLANs into Prime Core Minerals network builds in extra security into their network by virtually segregating their network so only devices on the same VLAN can connect to each other without going through the firewall. This allows the firewall to check and confirm if a device is permitted to view the other devices on the other VLAN.

Virtual segregation of networks also has benefits like:

- Improved performance
- Extra security
- Efficient management
- Controlled access
- Predictable network traffic

PCM VLAN Design

In a meeting with Prime Core Minerals managers, they confirmed that they have the following different devices either within their company or they want added to their network:

- Employee's – LAN and Wi-Fi
- Visitors (new setup)
- Printers
- Video Conferencing units
- CCTV
- IoT devices
- Servers
- Backups

From this list, our proposal is 10 VLANs and they are as follows:

Brisbane Office VLAN and DHCP overview		
VLAN Number	IP Range (.1 is gateway IP)	Name of VLAN
vlan10	192.168.10.0/24	Employee_LAN_Network
vlan11	192.168.11.0/24	Employee_WiFi_Network
vlan20	192.168.20.0/24	Visitor_Network
vlan30	192.168.30.0/24	Printer_Network
vlan40	192.168.40.0/24	Video_Conferencing_Network

vlan41	192.168.41.0/24	CCTV_Network
vlan50	192.168.50.0/24	IoT_Network
vlan90	192.168.90.0/24	Server_Network
vlan91	192.168.91.0/24	Backup_Network
vlan99	192.168.99.0/24	Network_Management

Site Office VLAN and DHCP overview		
VLAN Number	IP Range (.1 is gateway IP)	Name of VLAN
vlan10	192.168.110.0/24	Employee_LAN_Network
vlan11	192.168.111.0/24	Employee_WiFi_Network
vlan20	192.168.120.0/24	Visitor_Network
vlan30	192.168.130.0/24	Printer_Network
vlan40	192.168.140.0/24	Video_Conferencing_Network
vlan41	192.168.141.0/24	CCTV_Network
vlan50	192.168.150.0/24	IoT_Network
vlan90	192.168.190.0/24	Server_Network
vlan91	192.168.191.0/24	Backup_Network
vlan99	192.168.199.0/24	Network_Management

VLAN and Essential Eight Framework

By implementing VLANs onto Prime Core Minerals network and placing the management of the network equipment into a management VLAN (vlan99) and requesting users to log into them, we are abiding by the following Essential Eight Framework sections:

- Maturity Level 1
 - o Restrict Administrative Privileges

- Access is restricted and requires a different logon to the network hardware to view configurations.
- Admin access to network equipment is separated from other devices and users by being on a separate VLAN

VLAN and ISO27001

As VLANs are considered as another layer in network security, they are also mentioned in ISO27001:2022, sections:

- Annex A8.20 Network Security^{2, 3, 4}
 - o Network configurations are stored and secured in a central location and only users who are permitted access can access them
 - o Network is divided into separate and isolated VLANs and all cross VLAN communication must be verified and monitored through the firewall.
 - o All incoming/outgoing traffic is monitored through the firewall.
 - o IPsec IKEv2 encryption applied

This design also complies with ISO27001:2022 Annex A 8.22^{2, 3, 4}

- Network perimeters
 - o VLAN create a virtual separation of the network between corporate data, systems, backup network and high-risk zones like the visitor network

Any traffic transgressing to a different VLAN can only be approved by the firewall after it has inspected and verified it is permitted.

As the ISO27001 is a copyrighted document, AI was cross referenced with other sites, which it had also used in it's reply. The AI question and reply is in the reference section.

The VLAN implementation for this project has been covered in other technical artefacts.

References

<https://www.cyber.gov.au/business-government/asds-cyber-security-frameworks/essential-eight/essential-eight-maturity-model>

2 – <https://iso27001.com/iso-27001-annex-a-8-20-networks-security/>

3 – <https://www.isms.online/iso-27001/annex-a-2022/8-22-segregation-of-networks-2022/>

4 – Google Gemini

A new network design includes vlan and a user can only log onto the network equipment if they have the device local account and password and on the network. How does this comply with iso27001 standard. be specific and include exact section number and exact wording from iso27001 followed by this complies.

Your network architecture complies with **ISO/IEC 27001:2022** through three distinct Annex A controls. By enforcing a dedicated network path (VLAN restriction) paired with device-level local credentials, you satisfy rules around isolation, resource access, and fundamental configuration management. [1, 2, 3]

The specific clauses, exact standard wording, and alignment details are detailed below:

1. Network Segregation (VLAN Restriction)

- **Exact Section Number:** Annex A 8.22
- **Exact Wording:** *"Groups of information services, users and information systems shall be segregated in the organisation's networks."*
- **This complies** because your design uses a VLAN to partition the network infrastructure. Restricting equipment login access so that a user must physically or logically be "on the network" ensures that management traffic is segregated from general user or public traffic zones, directly preventing lateral movement. [1, 4]

2. General Network Security Management

- **Exact Section Number:** Annex A 8.20 [3]
- **Exact Wording:** *"Networks and network devices shall be secured, managed and controlled to protect information in systems and applications."*[4]
- **This complies** because your architecture explicitly protects the control plane of the equipment itself. Requiring a user to cross both a specific network boundary (the VLAN) and a local device challenge satisfies the baseline obligation to securely manage and control network entry points. [3]

3. Logical Access Control Rules

- **Exact Section Number:** Annex A 5.15 [5]
- **Exact Wording:** *"Rules to control physical and logical access to information and other associated assets shall be established and implemented based on business and information security requirements."* [6]
- **This complies** because you have instituted a strict multi-layered logical access rule: a valid credential set alone is insufficient if the request originates outside the designated management segment. This enforces a restricted logical pathway to sensitive assets. [7]

Important Audit Considerations

While your design creates a highly secure gate, ISO 27001 auditors will look for specific evidence during an evaluation:

- **The Risk of Local Accounts:** Relying on **device local accounts** instead of a centralized AAA server (like TACACS+ or RADIUS) makes tracking individual accountability difficult. Under **Annex A 5.16 (Identity Management)** and **Annex A 8.5 (Secure Authentication)**, you will need to prove how you manage, rotate, and revoke these local passwords when a network administrator leaves the company. [8, 9]

Would you like to explore how to map this design to **Annex A 8.5 (Secure Authentication)** to ensure your local password requirements (such as length or brute-force blocking) satisfy an auditor? [9]

[1] <https://www.isms.online>

[2] <https://www.isms.online>

[3] <https://www.isms.online>

[4] <https://consultantslikeus.co.uk>

[5] <https://www.isms.online>

[6] <https://www.youtube.com>

[7] <https://iso27001.com>

[8] <https://www.isms.online>

[9] <https://www.isms.online>